

Data Sufficiency

COMPLETE LEARNING GUIDE

What You'll Learn

Data sufficiency questions present a question and two (sometimes three) numbered statements, and ask NOT for the answer to the question but whether the given statement(s) provide ENOUGH information to answer it definitively — a subtle but critical shift in what's being evaluated. The standard answer choices test each statement's sufficiency independently before considering them together: statement I alone is sufficient; statement II alone is sufficient; both together are needed but neither alone suffices; either alone is sufficient; or the statements together are still not sufficient. The single biggest mistake candidates make is actually solving the underlying question using outside assumptions or 'typical' values instead of strictly checking whether the given information pins down one unique answer. This sub-module covers the standard answer-choice format, the correct testing order, numeric sufficiency examples, geometry-style sufficiency, combining statements, and the classic trap of assuming unstated 'reasonable' values.

1. The Standard Answer-Choice Format

Nearly all data sufficiency questions use a fixed answer template: (A) Statement I alone is sufficient, but II alone is not; (B) Statement II alone is sufficient, but I alone is not; (C) Both statements together are sufficient, but neither alone is; (D) Either statement alone is sufficient; (E) Even both statements together are not sufficient. Memorizing this exact structure lets you fill in a small A/B/C/D/E truth table as you test each statement, rather than reasoning about sufficiency from scratch each time.

2. The Correct Testing Order: Independence First

ALWAYS test Statement I completely alone (as if Statement II didn't exist), then test Statement II completely alone (as if Statement I didn't exist), and only THEN, if neither alone suffices, test them together — testing them together first is the most common procedural error, since it can make an insufficient single statement look sufficient by borrowing information it isn't entitled to use alone.

- Question: 'What is the value of x ?' Statement I: ' x is a positive integer less than 3.' Statement II: ' x is an even number.' Statement I alone: x could be 1 or 2 — not sufficient (two possible values). Statement II alone: x could be any even number — not sufficient. Together: x is a positive integer less than 3 AND even !' $x = 2$ uniquely. Answer: (C), both together are sufficient, but neither alone is.

3. Numeric Sufficiency: One Unique Value vs a Range

A statement is sufficient only if it narrows the answer to exactly ONE possible value (for a 'what is the value of' question) or definitively resolves a yes/no question in every case it allows — a statement that narrows the range but still leaves 2 or more valid possibilities is NOT sufficient, even if it feels close.

- Question: 'Is n an even number?' Statement I: ' n is divisible by 4.' Every multiple of 4 is even, so this ALWAYS answers 'yes' — sufficient, even though it doesn't tell you the exact value of n . A yes/no data sufficiency question only needs a definitive yes or definitive no in every case, not a unique numeric value.

4. Geometry-Style Sufficiency

For geometry questions ('what is the area of the triangle'), a statement is sufficient only if it pins down enough independent measurements or relationships to compute a single numeric answer — a statement establishing a shape is 'right-angled' without giving any side lengths, for instance, narrows the shape but does not by itself yield a numeric area.

- Question: 'What is the area of rectangle ABCD?' Statement I: 'The length is 8.' Not sufficient alone (no width). Statement II: 'The perimeter is 28.' Not sufficient alone (perimeter alone doesn't fix both length and width uniquely without more). Together: length = 8, perimeter = $2(8 + \text{width}) = 28$!' width = 6, area = 48. Answer: (C).

5. Recognizing 'Either Statement Alone Is Sufficient' (D)

Answer (D) applies when Statement I ALONE is enough to answer the question definitively, AND Statement II ALONE is also independently enough — these are two separate, self-contained sufficiency checks, not a case where the two statements say the same thing (they might use entirely different facts and both still work).

- Question: 'Is x positive?' Statement I: ' $x^2 = 9$ and x is odd' (this alone forces $x = 3$, since "3 is odd too — wait, need $x > 0$ test: actually if x is odd and $x^2 = 9$, $x = 3$ or $x = -3$, both odd, so NOT sufficient alone). A cleaner example: Statement I: ' $x = 5$.' Statement II: ' $x^2 = 25$.' Each alone fully determines $x = 5$, hence positive — answer (D).

6. Combining Statements Correctly (C) vs Neither Sufficing (E)

Only move to 'combine both' after confirming each fails alone. When combined, check whether the intersection of what each statement allows narrows to exactly one answer (!' C) or still leaves more than one possibility even together (!' E) — E is a valid, common answer, and questions deliberately include it as a trap for candidates who assume combining always helps.

- Question: 'What is the value of xy ?' Statement I: ' $x = 4$.' Statement II: ' $y > 0$.' Statement I alone: y unknown, not sufficient. Statement II alone: x unknown, not sufficient. Together: $x = 4$, y is any positive number — xy could be 4, 8, 40, anything positive. Still not a single value. Answer: (E).

7. The Trap of Assuming 'Reasonable' Outside Values

Data sufficiency strictly forbids using any information not explicitly given or standard mathematical/logical fact — assuming a 'typical' age, a 'usual' number of children, or that a variable is a positive integer when this isn't stated, are all invalid shortcuts that produce a wrong sufficiency verdict; if the statement doesn't restrict a variable's sign, parity, or type, all valid possibilities must be considered, including negative numbers, fractions, and zero where applicable.

Common Pitfalls

- Testing the two statements together before checking each one completely alone, which can make an insufficient statement look sufficient.
- Actually solving for the answer to the underlying question using outside 'reasonable' assumptions, instead of strictly checking sufficiency.
- Assuming a variable must be a positive integer when the statement never restricts its sign, parity, or type.
- Treating a yes/no question as needing a unique numeric value, when a statement that guarantees a definitive yes (or definitive no) in every case is already sufficient.
- Selecting (C) reflexively whenever neither statement works alone, without checking whether combining them still leaves multiple possible answers (which would be (E)).
- Confusing (D) — each statement independently sufficient — with statements that merely restate the same fact in different words.
- Forgetting to check that a geometry statement fixes ALL required independent measurements, not just some of them, before calling it sufficient.

Key Takeaways

- Data sufficiency asks whether the information is ENOUGH to answer definitively — never actually 'solve' using outside assumptions.
- Always test Statement I alone, then Statement II alone, and only combine them if both fail individually.
- For yes/no questions, a statement is sufficient if it guarantees the SAME answer (always yes, or always no) in every allowed case.
- For 'what is the value' questions, sufficiency means narrowing to exactly one possible value — a narrowed range with 2+ values is still insufficient.
- (E) — not sufficient even together — is a legitimate, commonly correct answer; don't assume combining statements always resolves the question.
- Never assume unstated properties (positive, integer, even) about a variable unless the statement or standard convention explicitly establishes it.

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